1. Clinical Question & Core Hypothesis

Clinical Question

Why do some tumors with high histologic immune infiltration fail to derive benefit from immunotherapy or show discordant biomarker signals? The Ontologist's tiers highlight DDR defects (HRD score) and MSH6/MSI as core modifiers of tumor immunogenicity, with liquid biopsy signals as complementary readouts. 2) Tier 1 paths used: I built on the HRD score path (CORE PATH 1) and the MSH6/MSI prognostic path (CORE PATH 2) because they provide mechanistic, and clinically actionable, axes linking DNA repair defects to neoantigen burden and immune engagement. I kept CTCs/EVs as a secondary modality to preserve hypothesis focus. 3) Tier 2 support: TP53-driven genomic instability and the EV-derived MDK/KRT17 signatures support mechanisms by which DDR status shapes tumor aggressiveness, antigen heterogeneity, and non-invasive detectability; AI-based biomarkers provide a translational route to integrate these signals. 4) Endpoint rationale: I selected a therapy_response proxy (immunotherapy response) because MSH6/MSI and DDR features have direct mechanistic and clinical implications for checkpoint inhibitor sensitivity, making therapy response the most problem-relevant endpoint. 5) Deliberate exclusions: I excluded CTCs/EV markers from the core two-anchor hypothesis to avoid exceeding the 2-anchor limit and to keep the mechanistic claim crisp; exploratory Tier 3 signals (PSA, pZVR, GLU/creatinine, PNI) were also excluded because they are indirect or cross-disease and would dilute the mechanistic focus?

Working Hypothesis

In tumors with high baseline TIL density, functional anti-tumor immunity during immune checkpoint blockade is contingent upon preserved antigen presentation and DDR status. Specifically, tumors harboring MSH6 deficiency together with a high HRD_score exhibit elevated neoantigen heterogeneity but concurrently upregulated immunosuppressive and antigen-presentation–defective programs. This leads to TILs that are abundant but functionally exhausted or ineffectual, resulting in poor immunotherapy response despite high TILs. Conversely, when antigen presentation capacity is intact (no B2M/HLA loss) and DDR-driven neoantigen load is high, high TIL density will more accurately predict favorable response. Therefore, a combined biomarker framework that integrates (1) TIL density (imaging/IF), (2) MSH6 status and HRD_score, and (3) antigen presentation integrity (B2M/HLA LOH, MHC I expression) will superiorly stratify patients for immunotherapy outcomes. Secondary liquid-biopsy signals (EV-derived MDK and KRT17) and AI-derived biomarker readouts will enhance non-invasive stratification, particularly for identifying cases with functional antigen presentation deficits not captured by tissue biopsy alone. Clinically, patients with high TILs but impaired antigen presentation may require DDR-targeted plus immune-modulatory combination strategies to convert: (a) high neoantigen burden with (b) preserved antigen presentation into meaningful checkpoint inhibitor responses.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	MSH6, HRD_score
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6/MSI prognostic path (CORE PATH 2) because they provide mechanistic, and clinically actionable, axes linking DNA repair defects to neoantigen burden and immune engagement.

HRD_score: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	TP53	Supporting modifier from knowledge graph
Secondary	CTCs	Supporting modifier from knowledge graph
Secondary	EV_mRNA	Supporting modifier from knowledge graph
Secondary	MDK	Supporting modifier from knowledge graph
Secondary	KRT17	Supporting modifier from knowledge graph
Secondary	AI-based biomarkers	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- TP53: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- CTCs: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.